



Mario Lavanga

Senior Biomedical Signal Processing & AI Engineer
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EDUCATION

KU LEUVEN

PHD IN ELECTRICAL ENGINEERING
Grad. September 2020 | Leuven, Belgium | Phd Graduation Video

POLITECNICO DI MILANO

MS IN BIOMEDICAL ENGINEERING
Grad. December 2015 | Milan, Italy | Final Grade: 100/100 cum laude

BS IN BIOMEDICAL ENGINEERING

Grad. December 2015 | Milan, Italy | Final Grade: 100/100 cum laude

ALTA SCUOLA POLITECNICA

DOUBLE DEGREE PROGRAMME WITH POLITECNICO DI TORINO
Grad. December 2015 | Turin, Italy | Final Grade: Excellent

LINKS

Mario Lavanga
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SKILLS

SIGNAL PROCESSING & ML

Real-time & embedded algorithms • Adaptive filtering • Wavelet / spectral • Time-series • Deep learning (CNN, Wide & Deep) • Causal inference (Granger, PCMCi) • Uncertainty quantification (VAE, SNPE) • Network physiology

PROGRAMMING & INFRA

Python • Matlab • C# / .NET • Bash • Tensorflow • Azure DevOps (CI/CD) • Docker • InfluxDB • Git

MEDICAL DEVICE & QUALITY

ISO 13485 QMS • IEC 62304 (SaMD) • FDA 510(k) • Design controls & verification • IP / patents

LANGUAGES

Italian | Native • English, German, French | Fluent • 汉语, Dutch | Basic

AI CODING TOOLS

Claude Code • Codex • Cline • Gemini CLI • MCP

ABOUT ME

I am a biomedical signal-processing and AI engineer. I build algorithms for vital-sign monitoring and ship them on regulated medical devices, both real-time on the device and in the cloud. My work runs across blood pressure and cardiovascular regulation (baroreflex, HRV), the brain (EEG, cerebral oximetry, NIRS) and the heart (ECG), carried from noisy clinical data to deployed, validated features while I lead projects across engineers and clinicians. Offstage, amateur theatre actor and language learner.

RESEARCH

HAMILTON MEDICAL AG | PROJECT LEADER

December 2024 –Present | Domat/Ems, Switzerland

Lead a signal-processing feature from clinical need to release into the cloud (SaMD) or embedded devices: requirements with ICU physicians, coordinating R&D engineers, verification, IP (patents), and documentation under our ISO 13485 quality system and IEC 62304 (FDA 510(k)).

HAMILTON MEDICAL AG | SENIOR RESEARCH ENGINEER

June 2022 –Present | Domat/Ems, Switzerland

Built real-time patient-ventilator asynchrony algorithms, deployed both *embedded on the flagship ventilator* and in the cloud. Added Wide & Deep networks for respiratory mechanics and effort, causal inference (PCMCi) for asynchrony, and an LLM ventilation-history summary for the bedside.

TNG - INS | POSTDOCTORAL RESEARCHER

Oct 2020 –June 2022 | Marseille, France

Modelled virtual brain aging (Digital Twin) with Prof. Viktor Jirsa and Prof. Svenja Caspers, using causal inference and uncertainty quantification (Bayesian VAE, SNPE) on 1,000 subjects (HPC). EU Human Brain Project; code on EBRAINS.

BIOMED GROUP - STADIUS | PHD FELLOW

Jan 2016 –Sep 2020 | Leuven, Belgium

Built a Perinatal Stress Calculator fusing ECG, HRV, EEG and respiration in noisy NICU data (UZ Leuven, with Prof. Van Huffel & Prof. Naeyaert). Earlier work modelled baroreflex sensitivity and arterial blood pressure. 8 first-author papers.

SELECTED PUBLICATIONS & CODE

M. Lavanga et al., The virtual aging brain: Causal inference supports interhemispheric differentiation in healthy aging. *NeuroImage*, 2023.

M. Lavanga et al., The role of the vagal nervous system in the recovery of blood-pressure control after cardiac arrest. *Physiol. Meas.*, 2017.

M. Lavanga et al., Maturation of the Autonomic Nervous System in Premature Infants via Heart Rate Variability. *Frontiers in Physiology*, 2021.

Open-source: in-browser 12-lead ECG classifier —1-D CNN on PhysioNet PTB-XL, ONNX, on-device (code).

Full list on Google Scholar: 624 citations, H-index 13, i10-index 18.

AWARDS AND VARIA

- Guest Editor, *Frontiers in Network Physiology* (2021-22) | invited talk at U. Virginia (Prof. Randall Moorman).
- Won competitive FWO research fellowship (2016) | KU Leuven Tech Transfer Laureate, Falling Walls Lab (2019).